

Amazon Data Analytics

Summary:

The Amazon Fresh Analytics project is an SQL-based database management and business analytics system developed for the e-commerce and retail domain. The project focuses on managing and analyzing large volumes of customer, product, supplier, order, and review data using MySQL Workbench. The main objective was to design an efficient relational database that supports business decision-making, improves customer satisfaction, and enhances operational performance.

The project includes database design through ER diagrams, identification of primary and foreign keys, and implementation of SQL concepts such as DDL, DML, constraints, joins, subqueries, aggregations, and normalization. Multiple SQL queries were developed to retrieve meaningful insights, including top customers, product sales performance, supplier analysis, revenue trends, and customer purchasing patterns.

Several datasets such as Customers, Products, Suppliers, Orders, Order Details, and Reviews were used to simulate a real-world e-commerce environment. The project also applied advanced database concepts like ACID transactions, constraints, and 3NF normalization to maintain data integrity and optimize database performance.

Overall, the project successfully demonstrated practical knowledge of SQL and relational database management systems. It improved skills in data modeling, query writing, business analytics, and database optimization while showing how SQL can be used to solve real-world business problems in the retail and e-commerce industry.